

AI TechStack 2025

Week 5 Task

September 18, 2025

1 Cat and Dog Classification

Objective

Develop a classical machine learning pipeline to classify images of cats and dogs using Histogram of Oriented Gradients (HOG) for feature extraction and three mandatory classifiers: Decision Tree (DT), Random Forest (RF), and Linear Regression (LR). Compare their performance using accuracy, confusion matrices, and learning curves. Optionally, incorporate additional classifiers to enhance the comparison.

Task Description

You are provided with a dataset of cat and dog images in the `'dataset'` directory, with filenames starting with `cat` or `dog` (e.g., `cat.1.jpg`, `dog.1.jpg`). Implement a pipeline to:

- Extract HOG features from grayscale images.
- Split the dataset into 80% training and 20% test sets, ensuring class balance (equal number of cats and dogs in each set).
- Train three classifiers: Decision Tree, Random Forest, and Linear Regression. For Linear Regression, adapt its continuous output to perform binary classification.
- Optionally, train additional classifiers of your choice to compare with the mandatory models.
- Evaluate and compare all models using accuracy, confusion matrices, and learning curves.
- Visualize sample results, including original images, HOG representations, and predicted labels for all classifiers.

Use `scikit-image` for HOG feature extraction, `scikit-learn` for classification, data splitting, and evaluation metrics, and `numpy-matplotlib` for data handling and visualization. Ensure the code supports adjustable parameters (e.g., HOG settings, train-test split ratio). Before implementing confusion matrices and learning curves, research their definitions and significance in model evaluation.

Steps to Follow

1. Data Preparation:

- Load images from the `'dataset'` directory.
- Assign labels (0 for cat, 1 for dog) based on filenames.

2. HOG Feature Extraction:

- Extract HOG features using `scikit-image` with adjustable parameters (e.g., `orientations=9`, `pixels_per_cell=(8,8)`).
- Store HOG features and visualizations for each image.

3. Train-Test Split:

- Split the dataset into 80% training and 20% test sets using stratified sampling to ensure class balance.
- Investigate how to maintain an equal number of cats and dogs in both sets and implement a solution if needed.

4. Classifier Training:

- Train a Decision Tree classifier on the training set's HOG features.
- Train a Random Forest classifier on the same features.
- Train a Linear Regression model and adapt its output for binary classification (e.g., by thresholding or another approach).
- Optionally, train additional classifiers (e.g., SVM, KNN) for comparison.

5. Evaluation and Visualization:

- Research the concepts of *confusion matrices* and *learning curves* to understand their role in model evaluation.
- Compute accuracy for all classifiers on the test set.
- Generate and visualize confusion matrices for each classifier to analyze prediction performance.
- Plot learning curves to assess model behavior with varying training data sizes.
- Visualize sample results from the test set, including original images, HOG representations, and predicted labels for all classifiers.

Notes on Linear Regression

Linear Regression typically produces continuous outputs, but this task requires its use for binary classification (cat vs. dog). You must devise a method to convert Linear Regression outputs into discrete class predictions. This is part of the challenge, and you are expected to explore and implement a suitable approach.

Improvements

Consider enhancements like:

- Adding other classifiers (e.g., SVM, KNN, Logistic Regression) to compare with the mandatory models.
- Tuning HOG parameters (e.g., `pixels_per_cell`, `cells_per_block`) for better feature extraction.
- Implementing data augmentation or preprocessing to improve model robustness.
- Experimenting with classifier hyperparameters to optimize performance.

Evaluation and Submission

Submissions are evaluated based on test set accuracy, quality of the train-test split (e.g., class balance), and effective use of confusion matrices and learning curves for comparison. Submit:

- Complete pipeline code (data loading, HOG extraction, train-test split, classification, and evaluation).
- Visualizations of sample results (original images, HOG representations, and predicted labels for all classifiers).

- Visualizations of confusion matrices and learning curves for all classifiers.
- Brief report detailing your approach, test accuracies, dataset split details, confusion matrix and learning curve interpretations, and any improvements or additional classifiers used.
- Summary of what you learned about Linear Regression, confusion matrices, and learning curves.